

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Pharm. Examination

University Theory Examination May - 2015

PH411 Pharmaceutical Analysis IV

Date: 05/05/2015, Tuesday

Time: 10:00 a.m. To 1:00 p.m.

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book is to be used.
3. Draw neat sketches wherever necessary.

SECTION I

- Q1** Attempt any TWO of the following:
- A** I. Explain the Principle of NMR Spectroscopy. 03
- II. Give a brief note on Shielding and Deshielding effect. 04
- III. Enumerate Applications of NMR Spectroscopy. 03
- B** Give the principle of Mass Spectroscopy. Draw well labeled diagram of double beam Mass spectrometer and explain function of each component. 10
- C** Write a Short Note On: 10
- I. Spin-Spin Coupling
- II. Detector Used in NMR Spectroscopy
- Q2** Attempt any TWO of the following:
- A** Write a detailed note on Fast Atom Bombardment Mass Spectroscopy (FAB-MASS) and MALDI Mass Spectroscopy. 10
- B** What is Chemical Shift? Write brief note on factors influencing chemical shift. 10
- C** Write a Short Note On: 10
- I. Metastable Ions or Peaks
- II. Coupling Constant

SECTION II

- Q 3** Attempt any TWO of the following:
- A** Discuss instrumentation of Atomic absorption spectrometer. Write a detailed note on radiation sources. 10
- B** Write note on interferences observed in atomic absorption spectrometers. 10
- C** Discuss Bragg's law. Write applications of X-ray diffraction. 10
- Q 4** Attempt any TWO of the following:
- A** Discuss characteristics of X-ray. Explain the principle of X-ray diffractometer. 10
- B** How supercritical fluid chromatography is advantages over other chromatographic technique? Discuss the basic principle of supercritical fluid chromatography. 10
- C** Write a Short Note On:
- I. Write about Pharmaceutical applications of atomic Spectroscopy. 05
- II. Discuss the basic principle of atomic emission spectrometry. 05
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